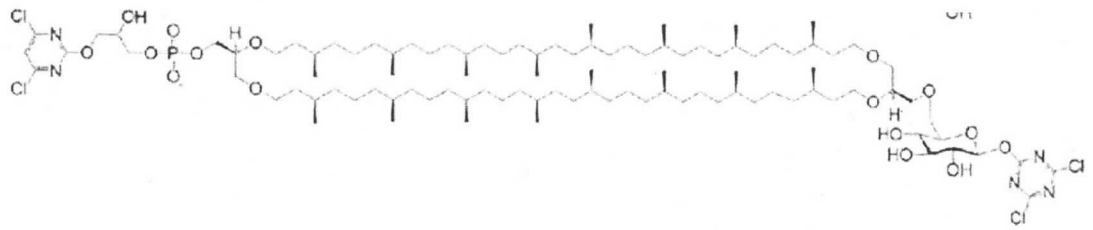
Biochemistry Cume 2-15-2019

1. You have isolated 2 new species of bacteria. Bacterial strain A grows best at 20° C, and bacterial strain B grows best at 40° C. Which strain would you expect to have a higher ratio of saturated to unsaturated fatty acids in its membrane? Explain your answer. (25 pts.)

2. Some microorganisms in the Archaea evolutionary branch thrive under very extreme conditions of low pH and high temperatures. In some of these species, their membrane lipid make-up is composed of the tetraether compound shown below:



A) How could a membrane lipid bilayer be constructed from this tetraether? (15 pts.)

B) Give 2 reasons why this archaeal compound provides thermal and chemical stability to the membrane under such harsh conditions.
Please justify your answer (20 pts.).

3. What structural features do a sphingomyelin and a phosphatidyl choline have in common? How do the structures of these two types of lipids differ? (15 pts.)

4. Write an equation, with structural formulas, for the saponification of a triacylglycerol. (25 pts.)